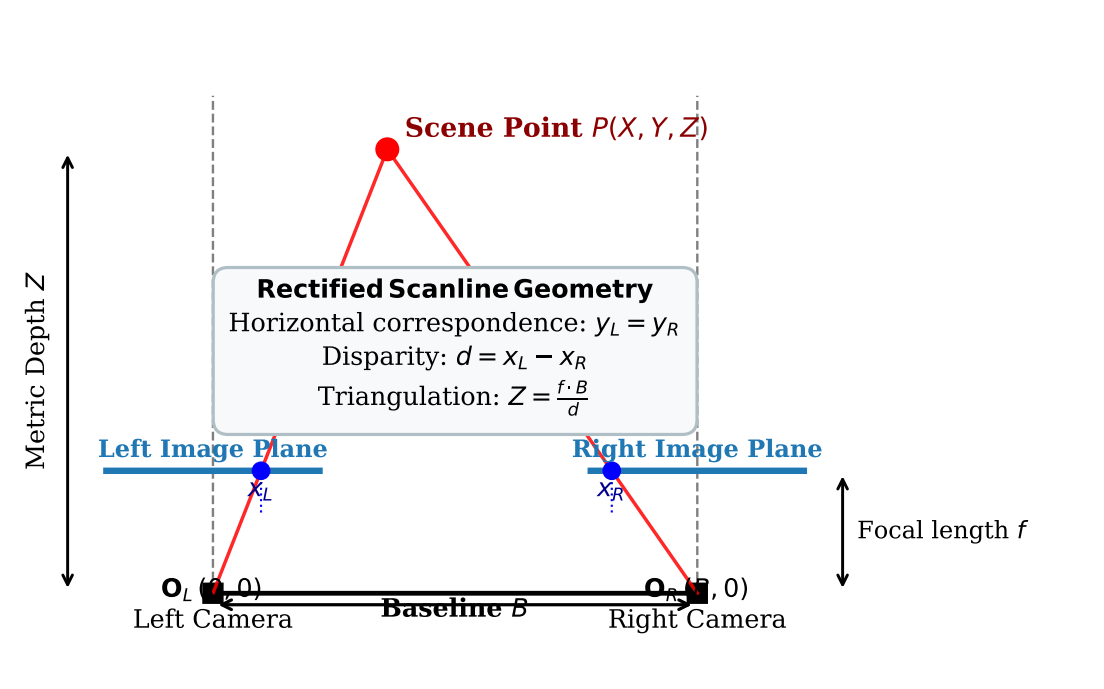
Scene Point $P(X, Y, Z)$
Rectified Scanline Geometry
Horizontal correspondence: $y_L = y_R$
Disparity: $d = x_L - x_R$
Triangulation: $Z = \frac{f \cdot B}{d}$
Metric Depth Z
Left Image Plane
Right Image Plane
x_L
x_R
Focal length f
$\mathbf{O}_L$ $(0, 0)$
$\mathbf{O}_R$ $(B, 0)$
Baseline B
Left Camera
Right Camera